

Data Flow Diagram

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	2 Marks

Data Flow Diagram

DFD Element Category	Shape / Symbol	Diagram Label	Description / System Role
External Entity	Oval / Rounded shape	User (Student)	Sends user actions (answers, subject choices) and receives generated quizzes, retention scores, and revision alerts.
External Entity	Oval / Rounded shape	Google Gemini API	Receives structured prompts from the app and returns generated test questions and explanations
Data Store	Solid Rectangle (no number)	User Database	Stores student credentials, selected topics, past quiz scores, and retention analytics.
Data Store	Solid Rectangle (no number)	Quiz / Item Bank	Holds raw topic prompts, active quiz questions, correct answer keys, and feedback text.
Process	Numbered Header Rectangle	1.0 User Authentication & Onboarding	Handles user login/signup, manages profile

			settings, and updates user account records.
Process	Numbered Header Rectangle	2.0 Adaptive Quiz Generation	Formulates API prompts, communicates with Gemini API, and writes generated tests to the quiz store.
Process	Numbered Header Rectangle	3.0 Assessment & Retention Engine	Scores quiz submissions, computes memory decay curves, and updates user performance history.
Process	Numbered Header Rectangle	4.0 Revision Schedule Predictor	Analyzes retention decay curves to calculate upcoming review dates and trigger study reminders.

Example

